

Portfolio

AI Scientist

Efficient Vision Generative Model

Multimodal Empathy Model

이아현 Ahhyeon Lee

M.S. in IT Engineering, Sookmyung Women's University (Seoul, Republic of Korea)

ahlee.sep@gmail.com | ah.lee@ivpl.sm.ac.kr

(+82) 010-9109-6271

<https://github.com/ahlee-smwu>

Contents

- 01** **Master's Thesis**
25.06~26.06
Few-Step Diffusion via Adaptive Prior
- 02** **Team Research Project I**
26.01~06
Multimodal Empathetic Response Generation
- 03** **Team Research Project II**
24.09~25.10
Multimodal Emotion Recognition and Empathy Modeling
- 04** **Personal Research**
24.09~25.05
Transformer Attention Minification

Master's Thesis (1/3)

Research Objectives

- Few-Step generating Diffusion
- Data and step adaptive prior in Diffusion

Motivation

- Problem Definition
 - Too slow sampling in Diffusion.
 - Diffusion's capacity is wasted in early time steps.
- Core Causes
 - Gaussian prior carries zero information
 - → model must learn structure from scratch.
 - → curved, entangled velocity trajectory.
 - → required many sampling steps.
- Related Works
 - Existing work optimizes the trajectory, not the inefficient prior that causes it.

Topic: Few-Step Diffusion via Adaptive Prior

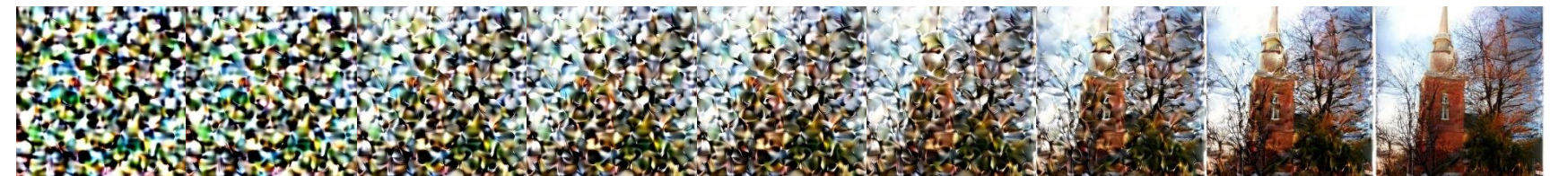
Title: RAPID: Robust Adaptive Prior Integration Diffusion

What is RAPID?

- Diagnoses sampling inefficiency at its source
 - The prior, not the trajectory
- Structured GMM prior, principled integration into diffusion
- Coarse-to-fine design → high-quality few-step generation
- RAPID = Robust Adaptive Prior Integration for Diffusion

Naïve Diffusion

250 steps, 0.63 img/s



RAPID

40 steps, 3.99 img/s



t = 0.0

t = 0.5

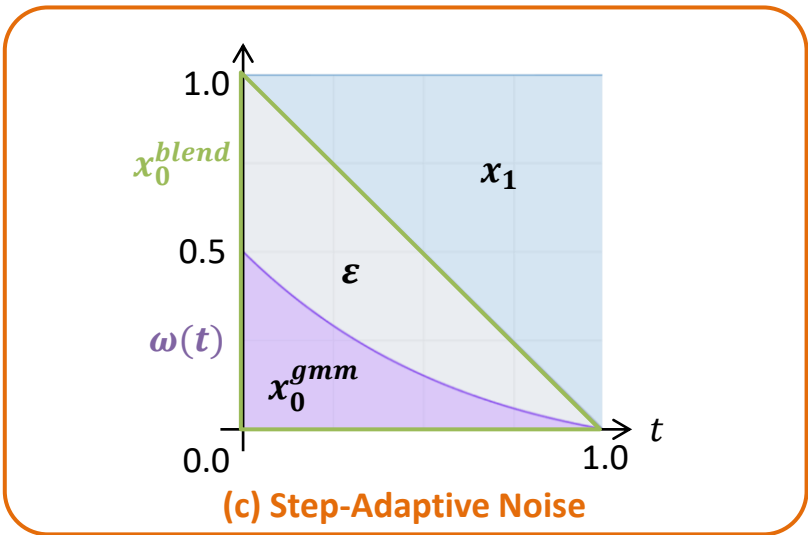
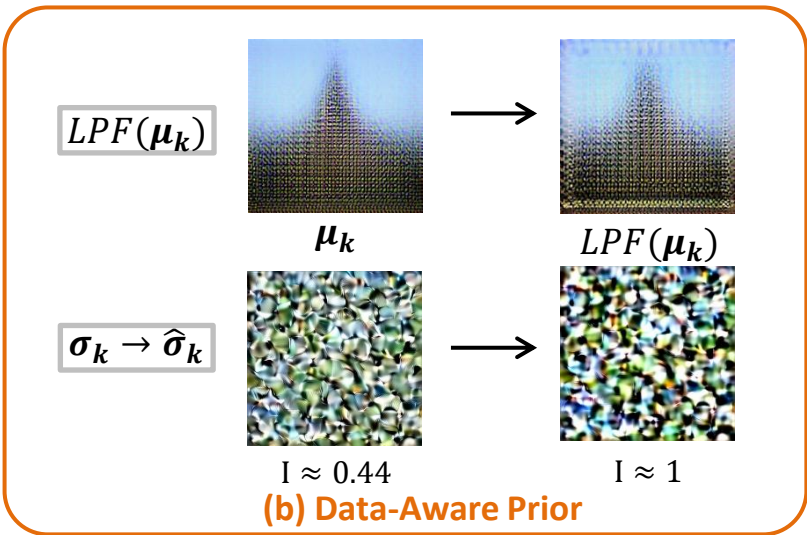
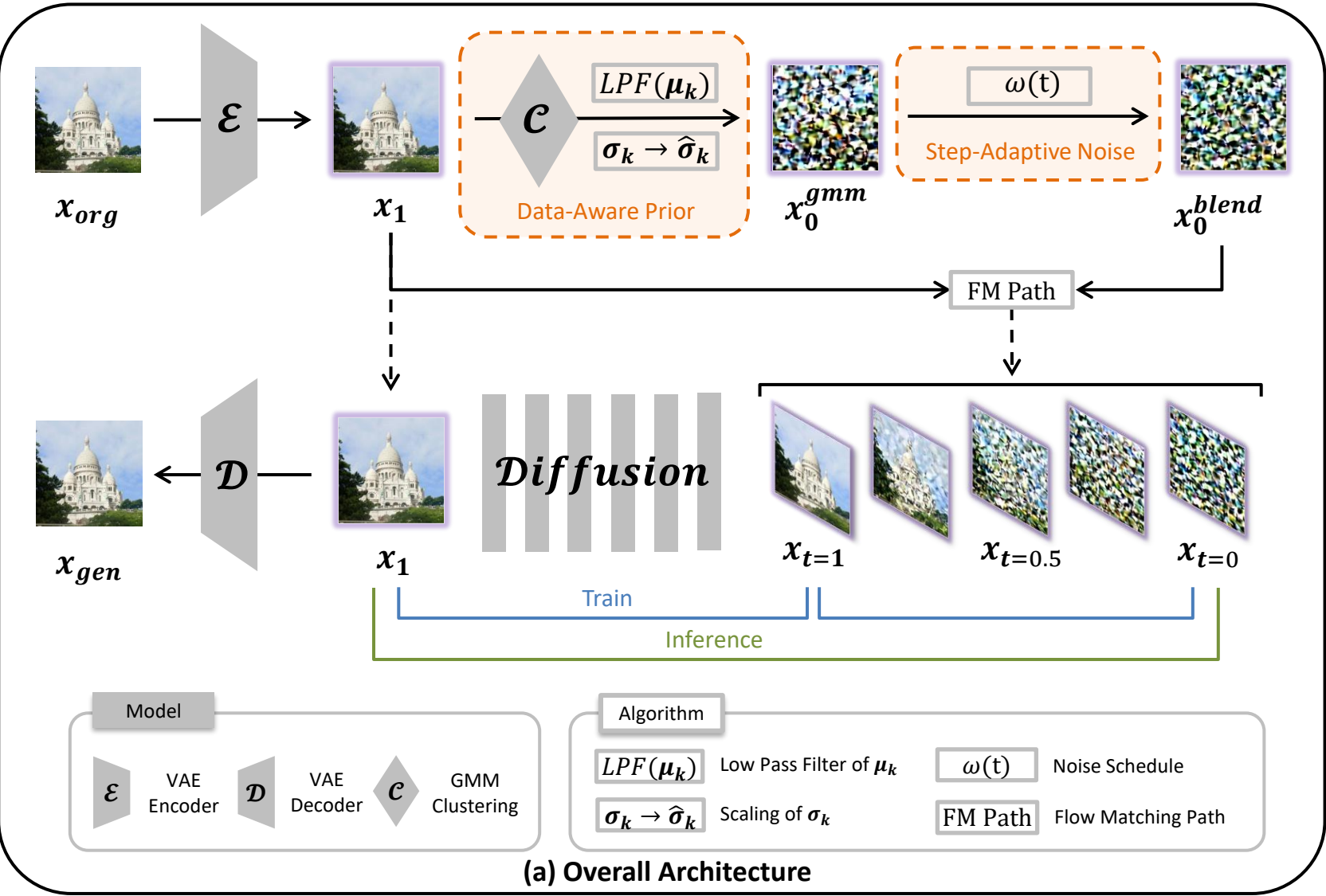
t = 0.75

t = 1.0

RAPID is 6.3x faster!

RAPID is 84% fewer steps!

Model Architecture



Method

➤ Data-Aware Prior

- GMM clustering of latent features
→ reveals structural modes already in target dataset.
- Cluster mean low-pass filter (LPF)
→ keeps coarse structure only.
- Cluster covariance normalization
→ restores diversity for diffusion's stochastic.

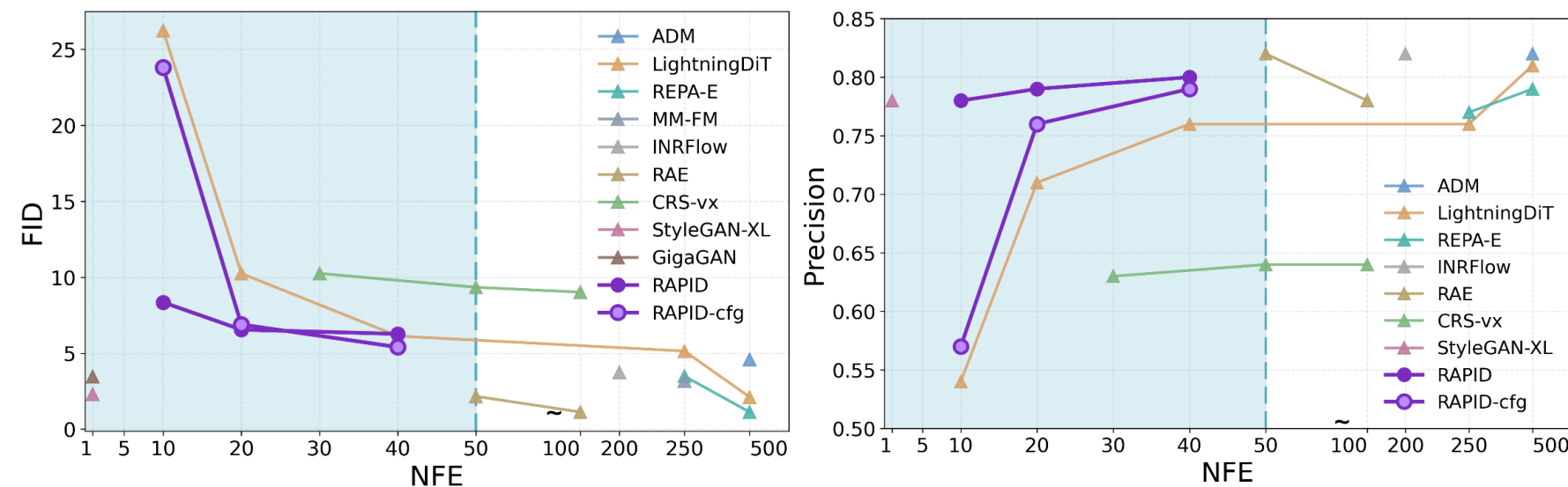
➤ Step-Adaptive Noise

- Coarse-to-fine principle
→ GMM prior's role should fade over time.
- SNR-aware blending $w(t)$
→ controls this decay.
- High $w(t)$ early (structure) → low $w(t)$ late (fine detail)

Experimental Results

➤ Quantitative Results

- Strong performance preserved at 20 to 40 NFE(= few sampling step).
- [40 NFE: FID 5.40 on ImageNet-1K](#), trained for 50 epochs
- [84% fewer NFEs, 6.3× faster sampling](#) than baseline



➤ Qualitative Results



Achievement

- The adaptive prior approach underlying this thesis was developed through two preceding studies ([First author](#)).
 - "An Optimized Diffusion Model Based on Adaptive Prior Distribution," KMMS Autumn Conference, Nov. 2025. ([Best Paper Award](#))
 - "Adaptive Prior Diffusion: Dataset-Aware GMM Prior for High-Fidelity Image Generation," KMMS Spring Conference, May 2026. ([Best Paper Award](#))
- The current thesis work is under review at *Pattern Recognition*.

Team Research Project I

Research Objectives

- Multimodal (Text/Audio/Video) empathy generation.
- Cross-attention for Speaker-Listener emotional cues.
- Emotion-driven empathy modeling.

Motivation

- Problem Definition
 - Text-centric models in empathy research.
 - Ignoring non-verbal cues.
- Motivation
 - Varying empathy by facial/vocal tones.
 - Need for Speaker's multimodal fusion related to Listener.

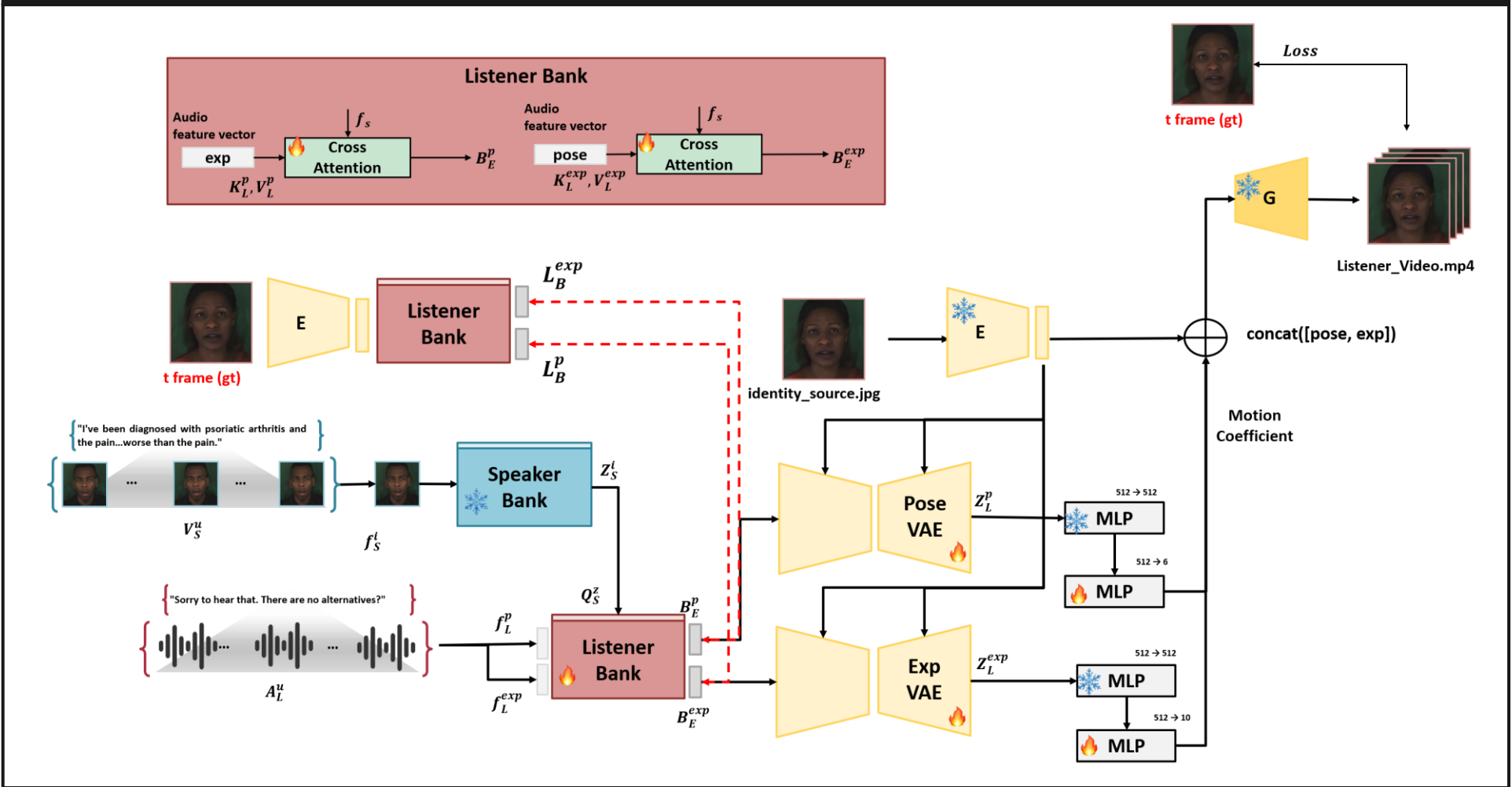
Achievements

- The current thesis work is under review at *AAAI 2027*.

Topic: Multimodal Empathetic Response Generation

Title: CAUSE: Sequential Empathetic Listener Reaction Generation from Response Speech

Method



Experimental Results



Team Research Project II

Research Objectives

- Multimodal emotion model via Dyadic datasets
- Vision & Audio integration for empathy.
- Multimodal alignment for consistency.

Motivation

- Problem Definition
 - Avatars limited to simple "talking head".
 - Lack of high-dimensional cognitive empathy.
- Motivation
 - Learning empathy correlations in Dyadic interactions.
 - Dynamic Video/Audio synchronization.

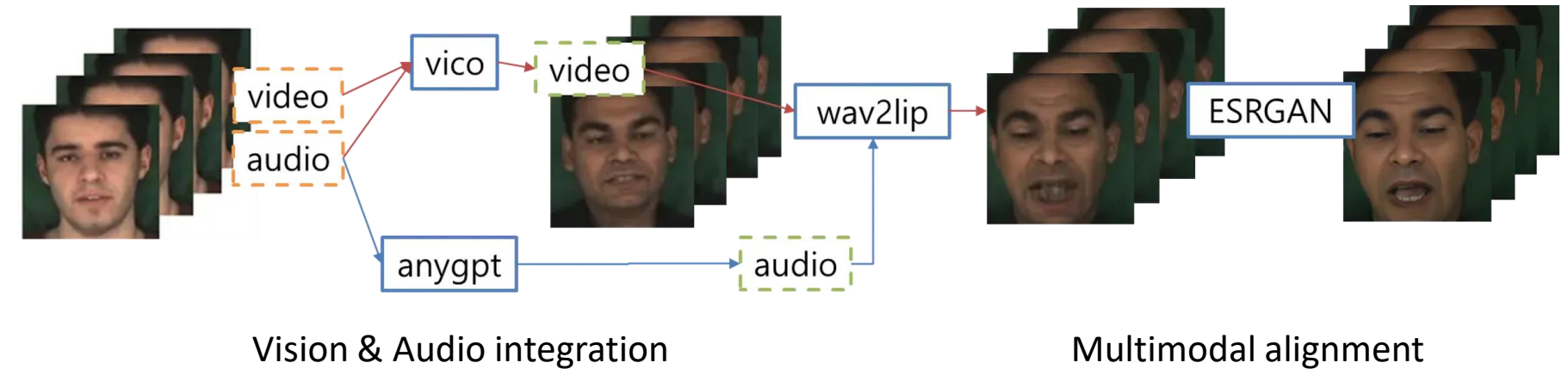
Achievements

- Published in *KMMS Autumn 2024* (First Author)
- Principal Investigator of “여대학(원)생 팀제 연구지원사업 (한국여성과학기술인육성재단)”
- Principal Investigator of “숙명여대 sw중심대학”

Topic: Multimodal Emotion Recognition and Empathy Modeling

Title: Multi-Signal-Based User Emotion Recognition and Cognitive Empathy Modeling

Method



Experimental Results

Speaker: neutral
Listener: happy



Speaker: neutral
Listener: worry



Speaker: neutral
Listener: furious



Personal Research

Topic: Transformer Attention Minification
Title: Attention Module Minification by Low-rank Factorization

Research Objectives

- Low-rank Factorization for Q, K, V matrices.
- Minimize FLOPs without accuracy loss.

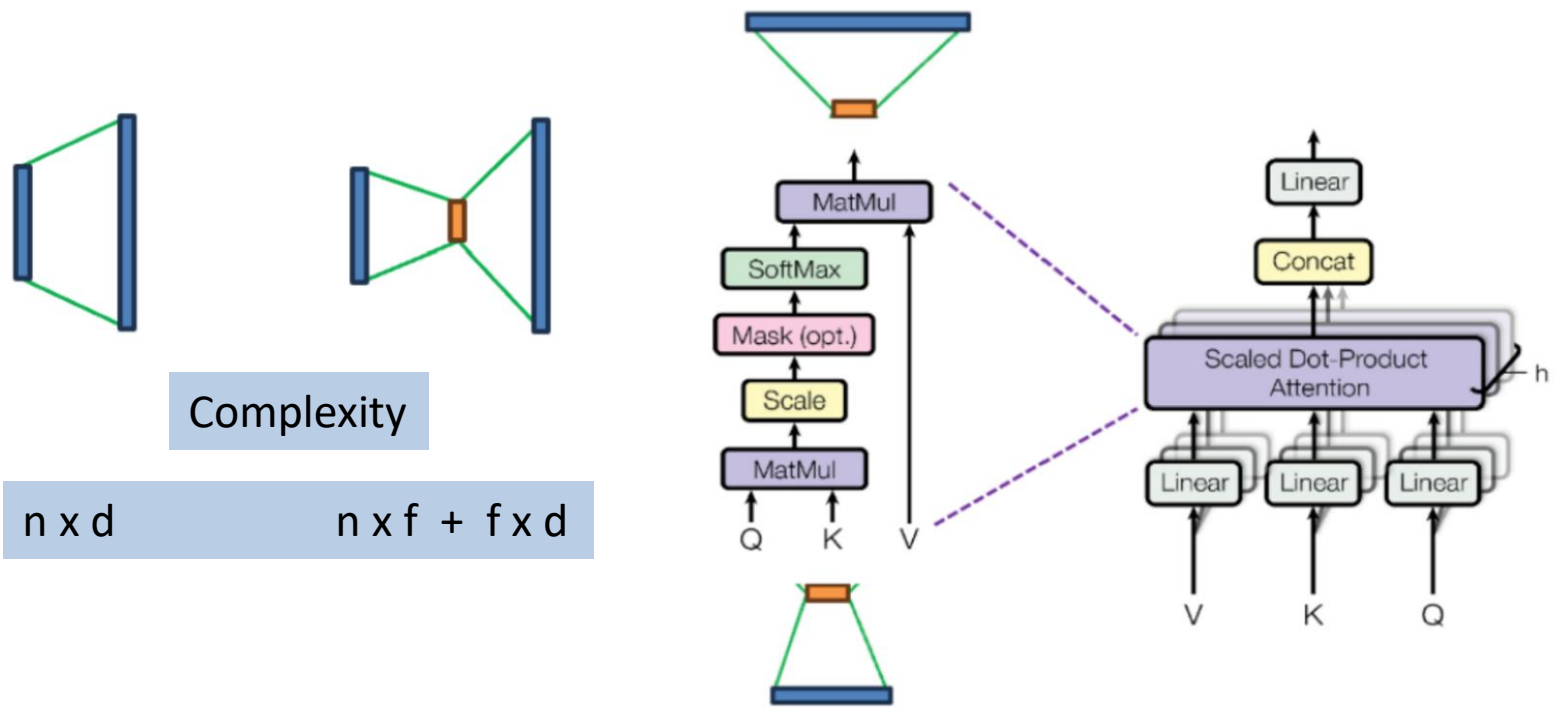
Motivation

- Problem Definition
 - Massive parameters and $O(n^2)$ complexity in matmul.
 - High computational overhead in Transformers.
- Motivation
 - Need for structural efficiency.
 - Maintaining performance with fewer calculations.

Achievements

- Published in *KMMS Spring 2025*
(First Author & Best Paper Award).

Method



Experimental Results

- Minification
 - 14–25% vs baseline
 - 92.2–94.2% vs BERT
- Preserved performance with +0.004–0.008 accuracy improvement

Research Achievement

AI Scientist

Efficient Vision Generative Model

Multimodal Empathy Model

이아현 Ahhyeon Lee

M.S. in IT Engineering, Sookmyung Women's University (Seoul, Republic of Korea)

ahlee.sep@gmail.com | ah.lee@ivpl.sm.ac.kr

(+82) 010-9109-6271

<https://github.com/ahlee-smwu>

Contents

- 01 Publications
- 02 Awards
- 03 Research Projects

Publications (1/3)

First Author

➤ International Journal (In Preparation/Under review)

▪ A. H. Lee and B. G. Kim*, "RAPID: Robust Adaptive Prior Integration for Diffusion", *Pattern Recognition* (Under review).

▪ A. H. Lee , Y. J. Cho and B. G. Kim*, "CAUSE: Sequential Empathetic Listener Reaction Generation from Response Speech", *AAAI 2027* (Under review).

➤ Domestic Conference

▪ A. H. Lee, H. S. Lee, and B. G. Kim*, “Adaptive Prior Diffusion: Dataset-Aware GMM Prior for High-Fidelity Image Generation”, *KMMS Spring Conference (KMMS)*, May. 2026 (**Best Paper Award**).

▪ A. H. Lee, Y. J. Cho and B. G. Kim*, “An Optimized Diffusion Model Based on Adaptive Prior Distribution” , *KMMS Autumn Conference (KMMS)*, Nov. 2025 (**Best Paper Award**).

▪ A. H. Lee and B. G. Kim*, “Attention Module Minification by Low-rank Factorization”, *KMMS Spring Conference (KMMS)*, May. 2025 (**Best Paper Award**).

▪ A. H. Lee, S. Y. Lee, T. K. Park and B. G. Kim*, “Development of Global Virtual Conferencing (GVC) Service Based on Real-time Translation Technique, *KMMS Spring Conference (KMMS)*, May. 2025.

▪ A. H. Lee, Q. U. A. Aisha, Y. J. Cho, N. R. Choi, S. I. Choi and B. G. Kim*, “Facial Emotion Recognition-Based Empathic Avatar Generation Technology”, *KMMS Autumn Conference (KMMS)*, Nov. 2024.

Publications (2/3)

Second Author

➤ International Conference

- Y. J. Cho, A. H. Lee, Byung-Gyu Kim*, and Jan Platos, “RMSF-ViT: Randomized Multi-Scale Fusion Vision Transformer”, **Communications in Computer and Information Science (CCIS) (Springer) , The 21th International Conference on Multimedia Information Technology and Applications (MITA), July. 2025 (Best Paper Award)**.
- Qurat Ul Ain Aisha, Ahhyeon Lee, Byung-Gyu Kim, and Jiwoo Kang, “Emotion Recognition Technique : A Comprehensive Review”, The Journal of Sciences and Innovation for Sustainable Peace, Vol. 3, No. 1, pp. 35-46, March. 2025.

➤ Domestic Conference

- Q. U. A. Aisha, A. H. Lee, and B. G. Kim*, “Temporal Gradient Consistency (TGC): Temporal Stability Assessment via Gradient Consistency for Video Stability” , KMMS Autumn Conference (KMMS), Nov. 2025 (Best Paper Award).
- Y. J. Cho, A. H. Lee and B. G. Kim*, “Frequency Separated Feature Conditioning for Diffusion-Based Video Super-Resolution”, KMMS Autumn Conference (KMMS), Nov. 2025.

Publications (3/3)

적응형 사전 분포 기반의 확산 모델 최적화 기법

이아현, 조유진, 김병규*
숙명여자대학교 IT공학과
e-mail : {ah.lee, yjcho}@ivpl.sm.ac.kr, bg.kim@sookmyung.ac.kr*

An Optimized Diffusion Model Based on Adaptive Prior Distribution

Ah-Hyeon Lee, Yu-Jin Cho, Byung-Gyu Kim*
Dept of IT Engineering, Sookmyung Women's University

요 약

확산 모델(Diffusion)은 stochastic한 특성으로 자유롭고 창의적인 결과를 우수하게 출력하여 많은 분야에서 주목받고 있다. 하지만 노이즈 분포와 데이터 분포를 직접 연결하는 모델링에 의해 많은 time step과 큰 model size가 필요하다는 한계가 있다. 특히 현행 연구의 사전(prior) 분포가 데이터와 무관하게 획일 표준 가우시안 분포($N(0, I)$)를 사용한다는 점이 이 한계의 원인이 된다. 따라서 본 연구에서는 Diffusion의 노이즈 분포, 즉 denoising의 시작점이 되는 사전(prior) 분포를 최적화하여 이 한계를 개선하고자 한다. 구체적으로 VAE 모듈의 latent 정보를 prior 분포를 추정 후 이를 반영하는 새로운 adaptive prior distribution 기법을 제안한다. 제안된 기법은 벤치마크 실험에서 메모리 등을 고려하여 FID score를 통해 generation task image control 역량을 검증하고 Adaptive prior distribution 연구의 중요성을 입증한다.

2025년도 한국멀티미디어학회 춘계학술발표대회 논문집 제28권 1호

저차원 행렬 분해를 통한 어텐션 연산 경량화

이아현, 김병규*
숙명여자대학교 IT공학과
e-mail : ah.lee@ivpl.sm.ac.kr, bg.kim@sookmyung.ac.kr*

Attention Module Minification by Low-rank Factorization

Ah-Hyeon Lee, Byung-Gyu Kim*
Dept of IT Engineering, Sookmyung Women's University

요 약

Transformer 구조는 자연어 처리와 컴퓨터 비전 등 다양한 분야에서 뛰어난 성능을 보이며 2017년 등장한 이후 꾸준히 주목받고 있다. 여기서 Attention 메커니즘은 정보 간의 관계를 효과적으로 모델링하는 핵심 구조로 알려져 있다. 그러나 Attention은 입력 길이에 따라 연산량과 메모리 사용량이 급격히 증가하는 한계가 있다. 특히 Attention 구조의 parameter 수에 비해 연산량인 FLOPs가 큰 특징이 있다. 따라서 본 연구에서는 Attention 연산을 저차원 행렬 분해(Low-Rank Factorization)를 통해 연산량을 크게 줄이고 모델의 효율성을 향상시키는 방법을 제안한다. 제안된 기법은 벤치마크 실험에서 연산량을 크게 줄이면서도 성능 저하가 없어 그 효율성을 입증한다.

2026년도 한국멀티미디어학회 춘계학술발표대회 논문집 제29권 1호

Adaptive Prior Diffusion: Dataset-Aware GMM Prior for High-Fidelity Image Generation

이아현, 김병규*
숙명여자대학교 IT공학과
e-mail : ah.lee@ivpl.sm.ac.kr, bg.kim@sookmyung.ac.kr*

Ah-Hyeon Lee, Byung-Gyu Kim*
Dept. of IT Engineering, Sookmyung Women's University

Abstract

A fundamental bottleneck of diffusion models lies in the standard isotropic Gaussian prior, which forces the model to traverse from uninformative noise to complex data distributions. We propose Adaptive Prior Diffusion (APD), which replaces the vanilla Gaussian prior with a dataset-adaptive prior constructed from GMM clustering of latent features. During training, the noise source x_0 is drawn from GMM component distributions $N(\mu_k, \Sigma_k)$; at inference, samples are drawn from the same components, maintaining consistency. On LSUN-Church-Outdoor, APD achieves Precision 0.730 with 40 steps, surpassing SOTA LDM-8 Precision of 0.640 (200 steps). While FID(13.973) does not yet reach SOTA levels, we show via velocity field analysis that this gap is a structural consequence of an inherent Precision-Recall trade-off, not a model capacity limitation – and this analysis is a core contribution.

RMSF-ViT: Randomized Multi-scale Fusion Vision Transformer

Yu-Jin Cho¹, Ah-Hyeon Lee¹, Byung-Gyu Kim^{1(✉)}, and Jan Platoš²

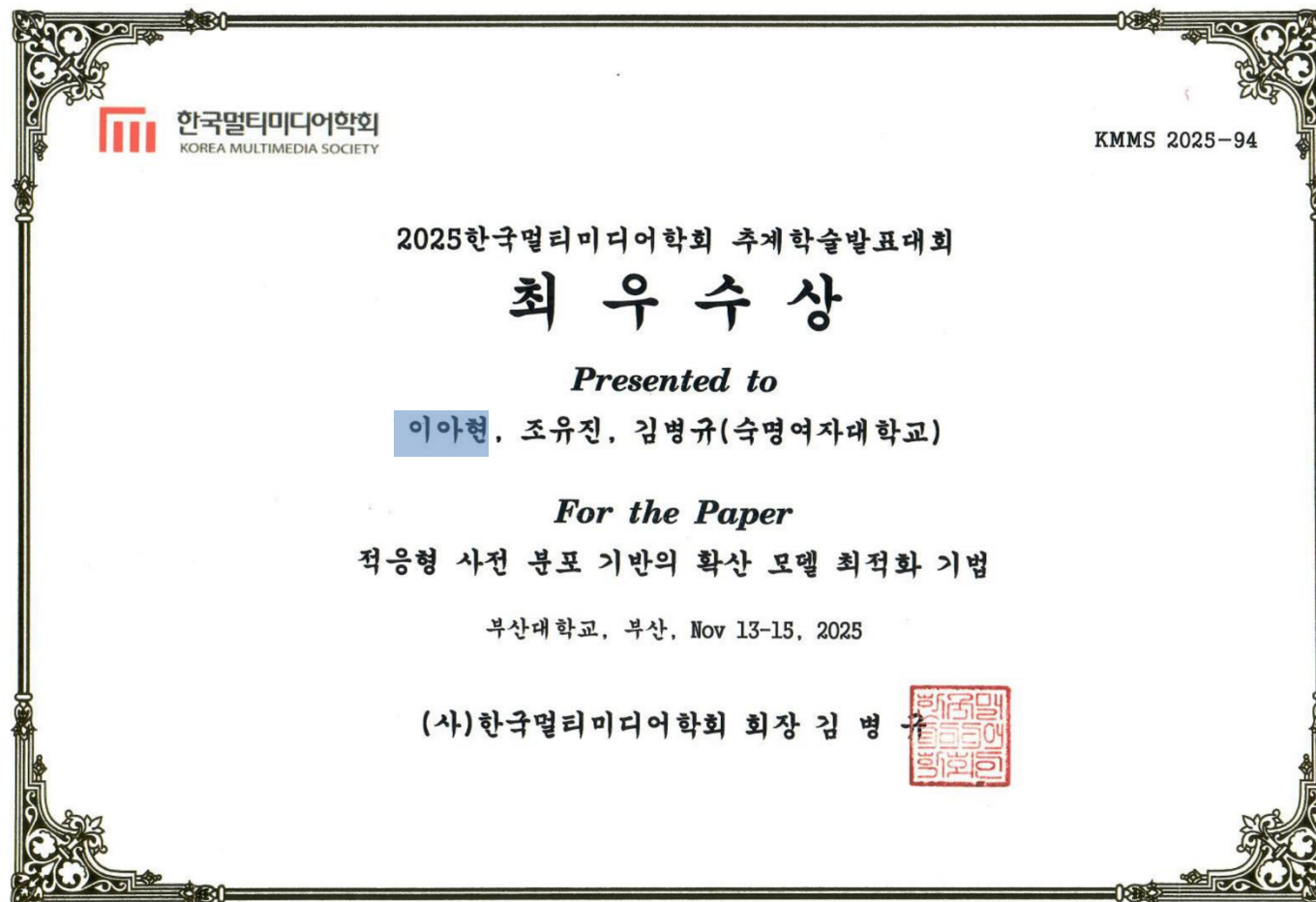
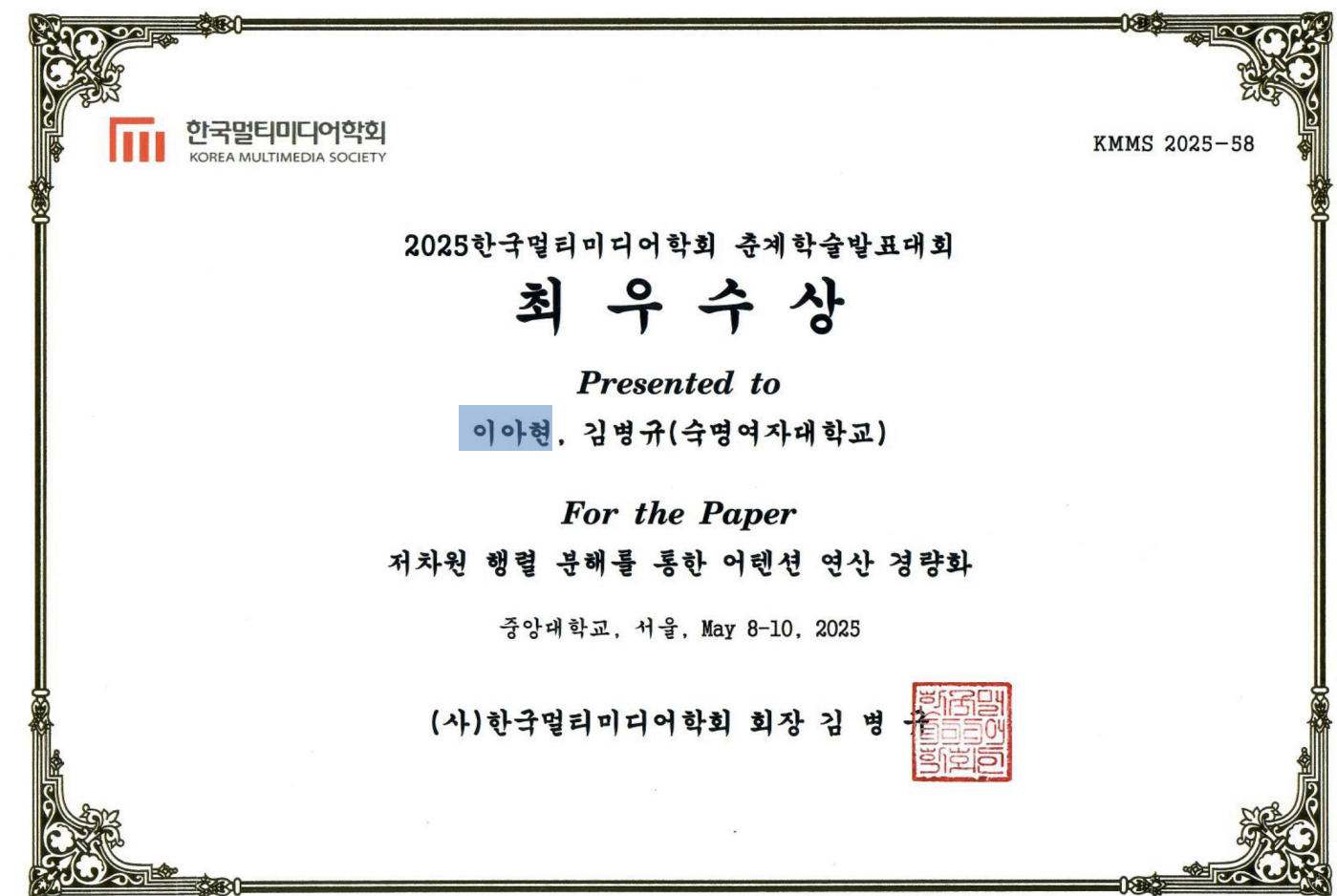
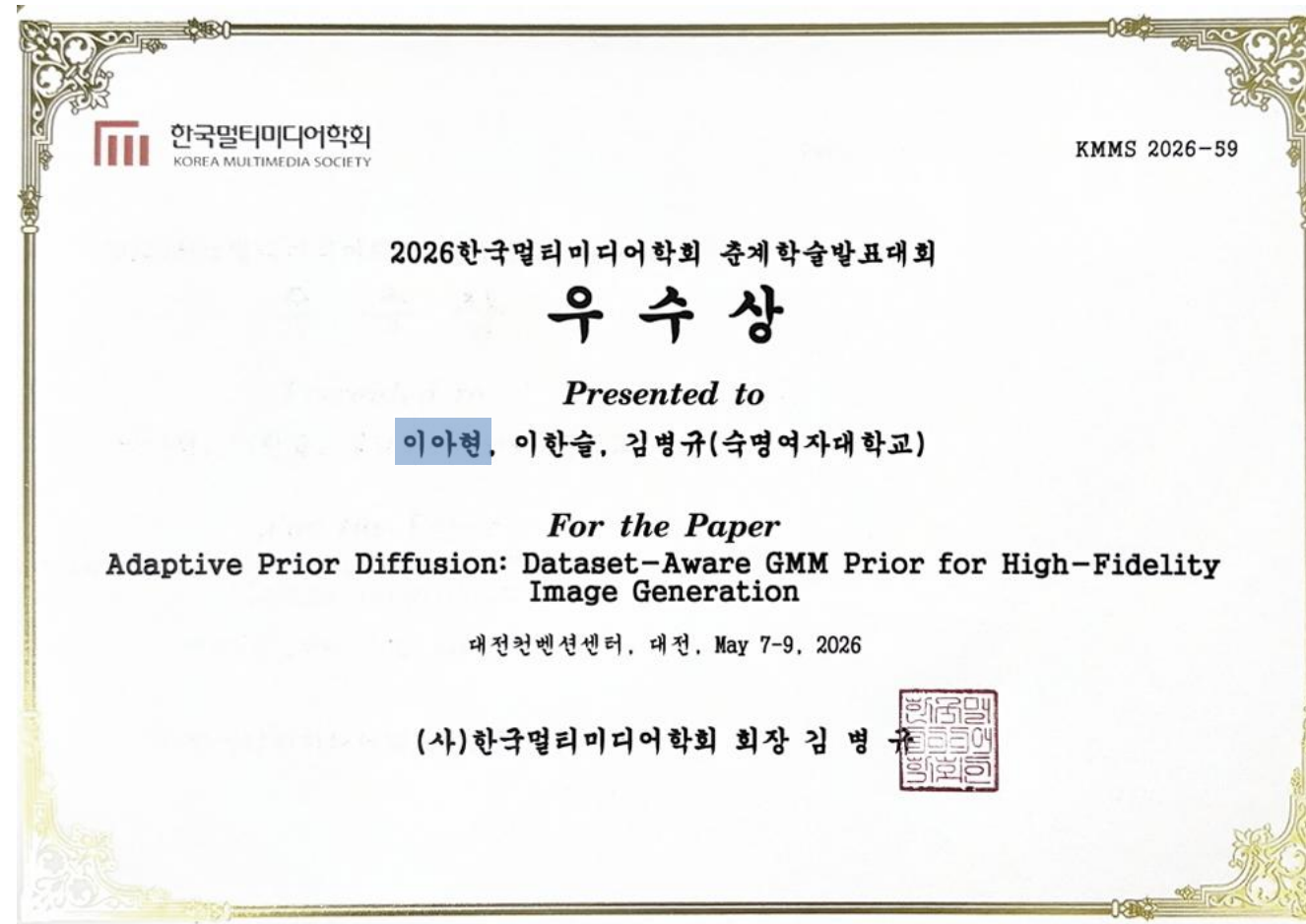
- ¹ Department of IT Engineering, Sookmyung Women's University, Seoul, Republic of Korea
{yjcho, ahlee}@ivpl.sm.ac.kr, bg.kim@sookmyung.ac.kr
² VSB – Technical University of Ostrava, Ostrava, Czech Republic
jan.platos@vsb.cz

Abstract. The Vision Transformer (ViT) has demonstrated remarkable performance in a wide range of computer vision tasks, such as image classification, object detection, and image generation. Unlike convolutional neural networks (CNNs), ViT benefits from a global receptive field, which enables more effective modeling of relationships between image patches. However, the lack of inductive biases makes ViT models difficult to train stably, especially on limited datasets. Without access to large-scale pre-trained weights, performance often degrades significantly. To address this issue, we propose a novel architecture called RMSF-ViT. It employs a progressive fusion strategy that incorporates fine-grained patch information beyond the fixed single patch size used in conventional ViT architectures. In addition, RMSF-ViT reduces the number of attention heads by half compared to vanilla ViT models. This design improves both performance and computational efficiency, as demonstrated on the CIFAR-10, CIFAR-100, Flowers, and Pets datasets.

Keywords: Vision Transformer · Multi-Scale Patch Embedding · Multi-Scale Fusion · Image Classification · Deep Learning



Awards



Research Projects

Period	Project	Funding	Role
2025.04–2025.10	Vision-based Emotion Recognition & Empathy Modeling (비전 신호 기반 사용자 감정인식 및 공감 모델링 연구)	Korea Foundation for Women In Science, Engineering and Technology (WISET) (한국여성과학기술인 육성재단)	Principal Investigator
2024.09–2024.11	Visual Signal Analysis and Empathy Modeling for User Emotion Recognition (사용자 감정 인식을 위한 시각 신호 분석과 공감 모델링 기술 개발)	Institute of Information & communications Technology Planning & Evaluation (IITP)	Principal Investigator
2026.03–2026.05	Probabilistic Empathy Response Generation via Multimodal Fusion (확률적 분포 기반 공감 반응 생성과 다중모달 신호 융합을 통한 사용자 공감 모델링 연구)	National Research Foundation of Korea (NRF) (한국연구재단)	Participant
2024.09–2026.08	Empathetic AI Women Engineering Education (공감형AI 여성공학인재양성 교육연구팀)	Ministry of Education (MOE) (4단계 BK21)	Participant
2024.03–2024.08	Generative AI-based Empathetic Avatar Generation (생성형 AI기반 공감형 아바타 생성기술 연구)	Institute of Information & communications Technology Planning & Evaluation (IITP)	
2024.09–2024.12	AV Coding & LF Media Technology for Tera-media (초실감 테라미디어를 위한 AV부호화 및 LF미디어 원천기술 개발)	Electronics and Telecommunications Research Institute (ETRI) (한국전자통신연구원)	Participant

Thank you for your attention.

이아현 Ahhyeon Lee

M.S. in IT Engineering, Sookmyung Women's University (Seoul, Republic of Korea)

ahlee.sep@gmail.com | ah.lee@ivpl.sm.ac.kr

(+82) 010-9109-6271

<https://github.com/ahlee-smwu>